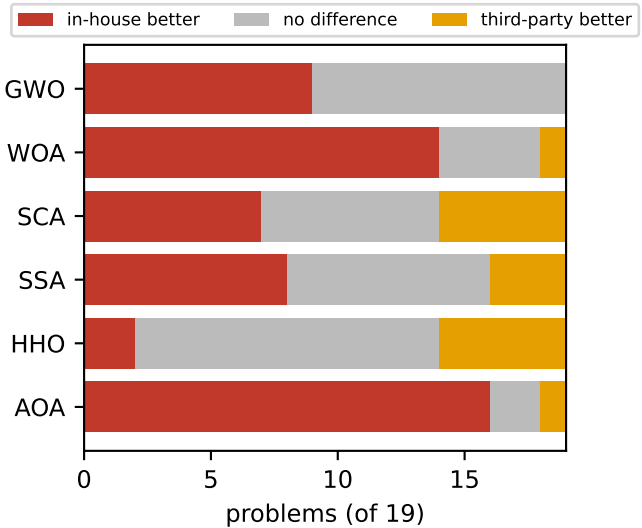
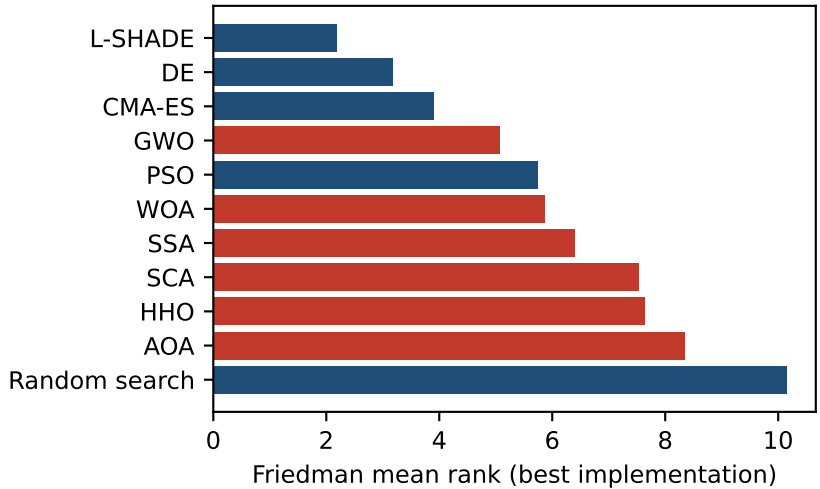

in-house better
no difference
third-party better
GWO
WOA
SCA
SSA
HHO
AOA
0
5
10
15
problems (of 19)


L-SHADE
DE
CMA-ES
GWO
PSO
WOA
SSA
SCA
HHO
AOA
Random search
0
2
4
6
8
10
Friedman mean rank (best implementation)